Unknowns, variables and solving mathematical statements

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1) Introducing variables and unknowns

Within mathematics, it is very much possible to have variables and unknowns be a part of a statement (i.e. equation or inequation).

Variables and **unknowns**, symbolized in mathematical expressions by non-numeric symbols of choice (which are often letters), are "mathematical entities" that when taking one or more specific values render the statement true. The two terms are often used interchangeably, but technically a variable may be regarded as an entity whose possible values within the statement value depends on the value of a second variable.

The presence of variable(s) or unknown(s) in a statement can be due to several reasons (they are described below), and normally, in these cases, it is necessary to inform the set of numbers (real, imaginary, etc) that the possible values of the variable or the unknown should belong to (i.e. the set of numbers in which the statement holds). When no specification is made, the set is assumed to be the real numbers.

In general, the statement containing a variable or unknown concerns a quantitative relationship that the variable or unknown is known for sure to satisfy (i.e., must respect). In cases where there is more than one unknow or variable, an statement establishes an algebraic relationship that their must satisfy conjointly. The values of the unknowns or variables involved in an statement that are able to hold it true are referred to as **solutions**. The procedure of finding the values the unknown can admit **and** hold true the statement is referred to as **solving** the statement (equation or inequation). It is only when a mathematical statement contains only one unknown that it makes sense to talk about solving the statement. This is because if there are more than one, the values that hold the statement true for one of them depend on the values of the other (which in turn is not pre-determined). Therefore, in this latter scenario, the statement might be regarded simply as what it is: a piece of information about the variables involved on it.

2) Solving mathematical statements (one unknown)

There are different methodologies to solve a statement containing one unknown rather than simply looking at it and trying to think hard based on the definition of operations contemplated in the mathematical statement – this can be quite difficult to do when the statement involves several different operations whose definition may not even be so intuitive. The one we are going to discuss here is what we refer to as the "**core procedure**", which does not involve anything other than logical thinking.

For the core procedure, it is necessary to realize that different mathematical statements can share the same solutions. **Different statements that share the exact same set of solutions can exist are said to be equivalent.** Realizing that statements are equivalent do not require knowing their solution: given two statements, by grasping the definition of the mathematical operations we have defined, it is possible to conclude that all values of x that render one statement is true must also render the other one true, necessarily. The definition of the code procedure goes as follow:

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The core procedure consists of writing a sequence of new statements that share precisely the same solutions (and only them) as the original statement and, at the same time, contemplate new statements whose solutions are progressively more intuitive to reason out. The new statement is obtained by inserting or performing new operations in the original one so that this new statement, despite being different from the original, is equivalent to it (i.e. their solutions are exactly the same).
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In the context of the core procedure, it's important to note that there is no strict methodological procedure, being it rather a manifestation of "creative logic" and intelligence. Finally, we note that when performing the core procedure, it is possible to use two or more newly developed statements that each carry a distinct subset of solutions of the original statement. In this case, the solving of the statement branches into solving these newly developed statements.

2.1) Solving equations: practical examples

The goal of this section is to do some examples to show how the core procedure for solving different equations works. The focus here is not on where the statement originates from, but simply to solve them.

Practical example 1:

$$2(x+2)^2 = 8 \tag{1}$$

The equation states that x must be a value that, when operated as indicated in its left hand side, results in a value of 8. Instead of trying to solve the puzzle directly by thinking out which value or values x can be, we can rather think about how we can write a new equation in a way that doesn't change the possible values of x can admit in the original one and at the same time makes it easier for us to figure out the solution.

We could think of dividing the whole equation by two: **if the value(s) of x must be such that $(x+2)$ squared multiplied by 2 is 8, then we must have that the value(s) of x is(are) also such that $(x+2)$ squared on its own is 4. Within the definition of multiplication, the latter premise is the only possible way that twice the value of $(x+2)$ squared can lead to 8. All solutions (and only them!) of the first clause are also necessarily solutions of the second clause within the scope of the multiplication operation.** We reach then:

$$(x+2)^2 = 4 \tag{2}$$

Further on, if the value of $(x+2)^2$ should be 4, then $(x+2)$ on its own may be 2. Indeed, the solutions of equation (3) below also solve equation (2).

$$x+2 = 2 \tag{3}$$

However, it is necessary to ask oneself: are the solutions of equation (3) the only ones that solve equation (2) within the scope of exponentiation? If you think about the definition of the square operation, there are two ways of obtaining a given positive number as output: the positive and negative input of the same magnitude. So, if $(x+2)^2$ is exactly 4, the result of $(x+2)$ can be 2 but also -2. So, equation (3) should be complemented by equation (4):

$$x+2 = -2 \tag{4}$$

Both, equations (3) and (4) are already in obvious formats. In other words, discovering their solutions is very easy: equations (3) and (4) can only be solved by 0 and -4, respectively, in the scope of the definition of summation as a mathematical operation.

Practical example 2:

$$\frac{(x-2)(x+2)}{(x-2)} = 4 \tag{5}$$

This is a very tricky example. As a consequence of the definition of multiplication and division, it is tempting to rush things without thinking carefully and state that diving and multiplying (or *vice versa*) a value by the same number keeps it unchanged. In other word, it is the same of never having operated the given value in the first place. This statement however is only true if the division is not by 0. As mentioned in the previous document, division is a mathematical operation that can lead to an undefined result (indetermination or non-existence), unlike summation, subtraction and multiplication. So, by thinking carefully, the right thing to say is that diving and multiplying (or *vice versa*) a value by the same number yields the same result provided that a division by 0 is not involved. This allows us to state equation (5) as equivalent to equation (6) **as long as $x \neq 2$.**

$$(x+2) = 4 \tag{6}$$

Interestingly, it is visible from the simplicity of equation (6) that, within the scope of the summation operation, its only solution is $x = 2$. Yet, this is the one condition in which equations (5) and (6) are not equivalent, as in fact $x = 2$ leads to an undefined result in the right hand side of equation (5) instead of a value of 4 as in equation (6). For **all** other values of x, equations (5) and (6) yield the same result. But since all other values of x in the left hand side of equations (5) and (6) lead to a number that is different than 4, there is no possible solution for equation (5). In other words, none of these other values renders the statement true.

2.2) Solving inequations: practical examples

Solving an inequation has the same equivalent principle as solving an equation, i.e., we aim to write a new inequation equivalent to the original one, which needs to be done mindfully within the scope of each mathematical operation involved in the statement.

Practical example 3:

$$\frac{-3(x-7)}{2} < 6 \tag{7}$$

Let's think about the modifications we can perform to generate new equivalent equations: if we multiply the two sides of the inequation by 2, would we be changing the solutions of the original inequation? Well, **if x must be numbers such that the resulting values of $-3(x-7)/2$ are smaller than 6, the double of these resulting values should be, each and all, smaller than 12 (think!).** So, indeed, the set of solutions is not altered: inequations (7) and (8) are equivalent.

$$-3(x-7) < 12 \tag{8}$$

Analogously, we could think now of dividing the two sides of the new inequation by 3. Indeed, upon reasoning within the definition of the division operation, it is possible to conclude that inequations (8) and (9) are equivalent (try doing it yourself!).

$$-(x-7) < 4 \tag{9}$$

A next tempting step is a subtraction of 4 in both sides of the statement. As stated in equation (9), x encapsulate all values such that when subtracted by 7 and multiplied by -1 leads to a value smaller than 4. In the realm of positive real numbers, all values of x for which $-(x-7)$ leads to a result ranging from 0 (inclusive) and 4 (not inclusive) are part of equation (9). The rest of the solution are all other values of x for which $-(x-7)$ equals 0 or a negative number, as zero and all negative numbers are also inferior to 4. Given that, if I want then to make a new statement that shares the precise same solutions of inequation (9) but that simultaneously has a left hand side that is always smaller than 0, I can write inequation (10):

$$-(x-7)-4 < 0 \tag{10}$$

Based on the previous reasoning, it is visible that inequations (9) and (10) are equivalent: all the solutions of inequation (9) also satisfy inequation (10). So far, we are good. Furthermore, by simply performing the operations indicated on the left hand side of inequation (10), we reach inequation (11):

$$-x+3 < 0 \tag{11}$$

A next idea could be a multiplication by -1. It is tempting to extrapolate the conclusion previously reached when assessing the viability of multiplying inequation (7) by 2 to multiplication in general (i.e. multiplication by any other number). But let's be careful. If we multiply or divide a number by -1, we simply change its sign, and the convention of which numbers are "larger than" swap for negative numbers relative to positive numbers. In other words, for positive numbers, a magnitude closer to 0 implies smaller, but, for negative numbers, a magnitude closer to 0 implies larger. This can be regarded as a convention established within mathematics: something we decided. Coming back to inequation (11), we then have that if x represents all values which multiplied by -1 and subtracted by 3 results into a negative number, each and all of these values of x must be such that when multiplied by -1, subtracted by 3 and multiplied **again** by -1 are positive numbers. In other words, inequations (11) and (12) share the precise same set of numbers as solution.

$$x - 3 > 0 \tag{12}$$

In inequation (12), we have that x are all values than when subtracted by 3 lead to a positive number. Given the algorithm of subtraction, these numbers must be only those larger than 3. No other set of numbers would satisfy equation (12). Thus, $x > 3$ is the solution of inequation (12).

Practical example 2:

$$\frac{(x+2)}{(x-3)} > 0 \tag{13}$$

Inequation (13) states that x is a value such that the result of $x + 2$ divided by the result $x - 3$ is a positive number. Within the scope of division, we know that there are two ways to get a positive number: either, both, the numerator and the denominator are (i) positive or (ii) negative. These are the only two ways to get a positive number as a result, and state that the same values of x that solve inequation (3) are those that satisfy simultaneously:[1]

$$(x+2) > 0 \,\&\, (x-3) > 0$$

or

$$(x+2) < 0 \,\&\, (x-3) < 0$$

To solve the first set of inequations – that is, find values of x that hold simultaneously both inequations true – we can simply solve them individually and evaluate the intersect of the solution set. Within the scope of subtraction and summation, we have that $x + 2 > 0$ holds true for all values of x that are larger than -2 while $x - 3 > 0$ holds true for all values of x that are larger than 3. Thus, the set of values that satisfy both simultaneously are $x > 3$. Similarly, for the second set of inequations, we have that $x + 2 < 0$ holds for all values of x smaller than

[1] We note that a set of inequations or equations that must be satisfied simultaneously can also be written within a "{" symbol in the language of mathematics.

-2 while $x - 3 < 0$ is satisfied by all values of x smaller than 3. Thus, the inequations that are part of the second set listed above can only be solved simultaneously by $x < -2$.

As discussed, solving any of the two sets of inequations above allow solving inequation (13), and so the solutions of inequation (13) are those that hold the sets true separately (the two sets cant (and shouldn't!) happen simultaneously). Ultimately, the values of x that solve equation (13) are the values smaller than -2 and the values larger than 3. In other words, values of x in between -2 and 3 are the only ones that do not solve inequation (13).

3) Solving of mathematical statements: final considerations

- It may be interesting to note that no matter how easy the equation or inequation ultimately looks when it carries one single operation at last, it is not always easy to ultimately figure out what should be the result for whatever operation, even upon grasping its algorithm. For example, in face of the equation $x^3 = 7.5$, it is not really obvious to see which should be the number that when multiplied by itself two times leads to a value of 7.5. So ultimately, depending on the complexity of the equation, a calculator might be the way out.

- **Every time a new equation or inequation is written, it must be checked if it shares the same solutions as the previous equation (and only have these shared solutions as solutions).** While not the topic of this book, it is interesting to note that solving of mathematical statements via the so-called core procedure shares fundamentally the same principle as coding. To write different parts of a python (or whatever language) code, it is necessary to make sure that each single part of the code that comes after the other is written in a way to do a given task does the intended task for all possible inputs that it can get from the output of the previous code part. Similar to solving equations, successfully coding requires a lot of mindfulness and intelligence !